08 · Case 4, part 3: her two acid-base tables

The words of this track, as it says them. Bold brown is the examiner.

Exam case 4, continued. Her two acid-base tables.

These two tables are starred, six marks each, and her 2024 deck fills in every cell. The trick is the order of each table's columns, because they run in opposite directions.

Six marks. Complete the table for a metabolic acid-base imbalance. How has blood pH changed? How have hydrogen ion levels changed? How does the bicarbonate buffer respond? What ultimately happens to carbon dioxide in the blood? And how does the breathing rate change as a result?

In the metabolic table, the hydrogen ions change first, and breathing responds.

Acidosis. The pH has fallen, because hydrogen ions in the blood have increased. The buffer gets rid of hydrogen ions by converting them into carbon dioxide and water, so carbon dioxide in the blood increases. The breathing rate increases, to get rid of that carbon dioxide.

Alkalosis. The pH has risen, because hydrogen ions have decreased. The buffer replaces the lost hydrogen ions by turning carbon dioxide and water into hydrogen ions and bicarbonate, so carbon dioxide in the blood decreases. The breathing rate decreases, to keep that carbon dioxide in.

Six marks. Complete the table for a respiratory acid-base imbalance. How has ventilation changed, hyper or hypo? How has that changed carbon dioxide in the blood? How does the bicarbonate buffer respond? What ultimately happens to hydrogen ions? And how does blood pH change as a result?

In the respiratory table, breathing changes first, and the pH follows.

Hyperventilation, which is Manaia. Carbon dioxide in the blood decreases. The buffer tries to replace the lost carbon dioxide by using up hydrogen ions and bicarbonate, so hydrogen ions decrease. Blood pH increases: alkalosis.

Hypoventilation, as after head trauma. Carbon dioxide in the blood increases. The buffer gets rid of the extra carbon dioxide by combining it with water and converting it into hydrogen ions and bicarbonate, so hydrogen ions increase. Blood pH decreases: acidosis.

Two marks. Compare the tables you completed. What do you notice?

Her answer, from the line under her table. They're very similar processes: the bicarbonate buffer system. And carbon dioxide and hydrogen ions are linked: whatever happens to one, happens to the other.

Once more. Both tables.

Metabolic acidosis: hydrogen ions up; the buffer converts them into carbon dioxide and water, so carbon dioxide rises, and breathing speeds up. Metabolic alkalosis: hydrogen ions

down; the buffer makes them from carbon dioxide and water, so carbon dioxide falls, and breathing slows. Hyperventilation: carbon dioxide down; the buffer uses up hydrogen ions, so they fall and pH rises: alkalosis. Hypoventilation: carbon dioxide up; the buffer makes hydrogen ions, so pH falls: acidosis. Compare them: it's the same buffer, and carbon dioxide and hydrogen ions always move together.